Finding and Assessing Research Articles with AI

ESS 111 Introduction to Health Careers. Library session worksheet and paper instructions.

Name	Date	Profession you are preparing for

How this assignment works

This assignment has two pieces that you turn in together.

- **The worksheet (Parts 1 to 6).** You fill this in during the library session. It shows how you found your articles, how you used AI to understand them, and how you checked the AI's work. Short answers are fine.
- **The paper (Part 7).** Two short write ups, one for each article, with APA citations and a bibliography. You will start it in class and finish it on your own.

Where you see a box, write in your own words. There are no trick questions. Honest observations about what worked and what did not are exactly what we are looking for.

Before you start

Sign in to **NotebookLM** (notebooklm.google.com) and **Gemini** (gemini.google.com) with your campus Google account.

Open **library.byui.edu** in another tab. You will search with **Library Search**, the main search box on the home page.

Never paste personal or identifiable health information about real people, such as patients you have observed, into any AI tool.

Part 1. How generative AI works

How an AI model is like a brain

Neurons and synapses. A large language model is a neural network. It is made of simple artificial neurons joined by billions of weighted connections, which work roughly like synapses.

Associations. Words are turned into numbers that sit close to related words. The model learns that therapist goes with patient, exercise, and recovery.

Learning. During training the model guesses the next word billions of times. Connections that lead to good guesses get stronger, much as practice strengthens synapses in your brain.

The limit. The model predicts what usually comes next. It has no way to know whether that prediction is true.

Not a calculator. A simple math program is deterministic. If A = 3 and B = 5, then C = A + B gives 8 every time you run it. An LLM is not. Ask it the same question twice and you get similar answers, but never exactly the same one.

1a. The prediction game. Write your prediction for each sentence as your instructor reads it.

Round	Sentence start	Your prediction	Did the class agree?
1	*Twinkle, twinkle, little star, how I wonder…*		
2	*The best treatment for low back pain is…*		
3	*I, Nephi, said unto my father…*		

> **From predicting to problem solving**
>
> **Then.** Early AI predicted an answer in one pass. Ask it for research articles and it wrote the most likely looking citations, and some of those articles did not exist.
>
> **Next.** Chat assistants were trained to follow instructions. They give a better answer in one try, sometimes after a quick web search, but they do not check their own work.
>
> **Now.** Reasoning agents like Gemini Deep Research work in a loop. They make a plan, search, read what they find, notice a dead end, change their search terms, and try again.
>
> **Picture a maze.** A prediction model draws the whole path in one guess. An agent walks it, backs up from dead ends, and tries another branch.
>
> **Still predicting.** Underneath, every step is still prediction. The loop makes AI more persistent, not more truthful, so you still trace claims back to the original.

1b. In your own words, how does a tool like Gemini produce an answer? What did round 3 teach you about asking AI for research articles? Why do you still need to check the work of an agent like Deep Research?

1c. The sliding scale of subjectivity. The more subjective a question is, the more critical you need to be of the answer. Use the scale on the resource pages at the end. Add two questions about your future profession. Place each on the scale and say how you would check the answer.

A question you might ask AI	Where on the scale?	How would you check the answer?
What muscles make up the rotator cuff?	*Objective*	*Compare with my anatomy textbook. Low risk, but still check any numbers.*

A question you might ask AI	Where on the scale?	How would you check the answer?

Part 2. Find two original research articles

Original research means the authors collected or analyzed their own data and report what they found. It usually has sections called Methods and Results. A review, meta analysis, editorial, or news story is not original research, even if it appears in a scholarly journal.

Gemini and Library Search each do one job here. **Gemini builds your search strings. Library Search finds the real articles.** Library Search is the main search box on library.byui.edu, and it searches most of the library's scholarly databases at once. Every article it returns actually exists, which is never guaranteed when an AI names an article.

Let Gemini interview you to build search strings

When a prompt is vague, AI fills the gaps with guesses. Having it interview you first gives you better results. Paste this into Gemini and answer its questions.

> **Prompt 1. The search string interview (Gemini)**
>
> I am a first year college student preparing to become a [YOUR PROFESSION, for example physical therapist assistant]. I need to find two original research articles that would help me in that profession. I will search for them in Library Search, my university library's main search box, which searches many scholarly databases at once.
>
> Before you suggest anything, interview me. Ask me one question at a time about the kind of patients or clients I want to work with, the problems I am curious about, and what I already know. Stop after no more than five questions.
>
> Then give me three search strings I can paste straight into Library Search. In each string, put exact phrases in quotation marks, group synonyms with OR inside parentheses, and join two or three different ideas with AND. One idea can be a word like trial OR participants, which helps find original research instead of reviews. Explain each string in one sentence. Do not suggest specific article titles, because you may invent articles that do not exist.

Search with Library Search

1. Go to **library.byui.edu** and paste one of Gemini's search strings into the **Library Search** box on the home page.
2. On the results page, check the **Peer Reviewed** and **Full Text** limiters. The peer reviewed box may be labeled Scholarly.
3. Consider limiting to the last ten years so the research is current.
4. If the results look wrong, such as thousands of hits, almost none, or mostly reviews, tell Gemini what happened in the same chat and ask for a revised string.

5. Read titles and abstracts. Look for words like *participants*, *randomized*, *trial*, *we measured*, or *we recruited*. Skip anything with *review* or *meta analysis* in the title.
6. Open the full text and check that it has Methods and Results sections. Download the PDF of each article you choose. Keep the article record open, because you will use its **Cite** button in Part 5.

2a. Record your two articles.

	Article title	Journal and year	How do you know it is original research?
1			
2			

2b. Copy the search string that worked best in Library Search. Did any of Gemini's suggestions lead you astray?

Part 3. NotebookLM or Gemini?

Both tools come from Google, and both use large language models. The difference is what they work from.

	NotebookLM	Gemini
What it works from	Only the sources you upload, such as your two article PDFs	Its general training plus web searches
Strengths	Answers are tied to your sources, with numbered citations you can click to see the exact passage. Good for summarizing, comparing two articles, and finding the Methods and Results	Knows a little about almost everything. Good for building search strings for Library Search, explaining unfamiliar words like randomized controlled trial, checking APA formatting, and citing web pages
Weaknesses	Knows nothing beyond your sources. Can still misread or oversimplify a study. Its summaries often reuse the article's wording, which raises your Turnitin score	Much more likely to invent facts, statistics, and even whole articles that do not exist. Sounds just as confident when it is wrong
Use it when	You have the article and want to understand what it says	You do not have a source yet, or you need general help

3a. For each task, write which tool you would use and why in a few words.

Task	NotebookLM or Gemini?	Why?
Explain what a *control group* is		
Tell me how many participants were in my article		
Suggest keywords for a new search		
Compare the findings of my two articles		

Part 4. Rapidly assess your articles with NotebookLM

1. In NotebookLM, create a new notebook and click **Add source**. Upload the PDFs of both articles.
2. Paste Prompt 2 into the chat, once for each article. Replace the brackets with your details.
3. Click the numbered citations in NotebookLM's answer to see the passage it used.

> **Prompt 2. The rapid assessment (NotebookLM)**
>
> Using only the source [ARTICLE TITLE], answer these four questions. Keep each answer to two or three sentences and cite the passage you used.
>
> **1. Research type.** Is this original research, where the authors collected or analyzed their own data? Or is it a review, a meta analysis, an editorial, or something else? What in the article tells you?
>
> **2. What the study did.** Who were the participants, how many were there, and what did the researchers do or measure?
>
> **3. What the study found.** What were the main results?
>
> **4. Relevance.** How could these findings matter to a future [YOUR PROFESSION]?
>
> If the article does not answer one of these questions, say so instead of guessing.

AI tools sound equally confident whether they are right or wrong. Next have the AI check itself, then check it yourself.

> **Prompt 3. The self check**
>
> Now check your own work. List every number and factual claim in your last answer. For each one, quote the exact sentence from the article that supports it and tell me which section it comes from, such as Methods or Results. Rate your confidence as high, medium, or low. Flag anything you cannot find in the article and correct any mistakes.

4a. Pick two claims from NotebookLM's answers, one from each article, and find them in the article yourself.

Claim NotebookLM made	What the article actually says (section and page)	Match?
Article 1		
Article 2		

4b. Did NotebookLM get anything wrong, leave anything out, or overstate a finding? Did the self check catch it?

Part 5. Cite your sources in APA style

Articles from Library Search. Use the cite tool

Library Search creates citations for you. Open the article record, click **Cite**, choose **APA 7th edition**, and copy the reference. These tools are fast but not perfect, so check the result against the checklist below or have Gemini check the formatting with Prompt 4.

- ☐ Authors are listed by last name and initials, in the order shown on the article.
- ☐ The article title is in sentence case. Only the first word, the first word after a colon, and proper nouns are capitalized.
- ☐ The journal name is in title case and *italics*. The volume number is also in *italics*.
- ☐ The DOI is written as a web address that starts with https://doi.org/ and it opens the right article.

Prompt 4. Check a library citation (Gemini)

Check this APA 7th edition reference for a journal article and tell me about any formatting errors. Look at author names and initials, the year, sentence case in the article title, title case and italics for the journal name, italics for the volume, the issue and page numbers, and the DOI written as https://doi.org/ followed by the number.

Do not change any facts such as names, dates, titles, or numbers. If a fact looks wrong, tell me to check it against the article instead of fixing it yourself.

Here is the reference. [PASTE THE REFERENCE FROM THE LIBRARY CITE TOOL]

Web sources with no cite tool. Use AI, but give it the details

Web pages from sources like professional associations or government agencies have no cite button. AI can format these quickly, but when you ask it to recall a source it often invents authors, dates, and web addresses. The safe approach is to give the AI the details yourself and ask it only to format them. NotebookLM also works if you add the web page as a source.

> **Prompt 5. Cite a web page (Gemini or NotebookLM)**
>
> Create an APA 7th edition reference list entry and a matching in text citation for the web page below. Use only the details I give you. Do not invent an author, date, or web address. If something APA requires is missing, tell me what it is instead of guessing.
>
> Source details. Page title [ ]. Author or organization [ ]. Publication or last updated date, or write none [ ]. Website name [ ]. Web address [ ].

Citing your AI use

Your paper will include a short AI use statement and APA references for the AI tools you used. Use these patterns and replace the bracketed parts.

- Google. ([Year, Month Day]). *[Short title describing your notebook]* [NotebookLM chat]. NotebookLM. https://notebooklm.google.com
- Google. ([Year, Month Day]). *[Short title describing your chat]* [Google Gemini chat]. Gemini. [Share link, if your account allows one]

APA guidance on citing AI is still changing. If your professor or the Library Citation Guide Videos show a different format, follow them.

5a. Write the APA reference for each of your articles. Check each one against the checklist.

Article 1

Article 2

5b. Practice. Find one web page from a professional association in your field, such as the American Physical Therapy Association, and use Prompt 5 to cite it.

Web page reference

Part 6. Meet EndNote Fusion

EndNote Fusion (formerly called RefWorks) is a free citation manager from the library. It saves your sources in one place and adds in text citations and a bibliography to your paper as you write, through an add in for Word or Google Docs. With only two sources this paper does not need it, but you will be glad to have it set up when you write longer papers with ten or twenty sources.

6a. Check off what you completed. Setup steps are on the resource pages.

☐ Created an EndNote Fusion account with my campus email

☐ Saved at least one of my articles to EndNote Fusion

☐ Installed the EndNote Fusion add in for Word or Google Docs (optional today)

Part 7. The paper

Now put it all together. The requirements below come from the instructor.

Requirements

- Write about the **two original research articles** you found in Part 2 that would help you in your chosen profession.
- For each article, write **one or two paragraphs of 150 to 350 words** that answer three questions. Why did you choose this article? What did the study do and find? How will the information help you?
- **No quotations.** Describe the articles in your own words. Points will be deducted if quotations are used.
- Put an **APA in text citation** at the end of the paragraphs about each article, for example (Author & Author, Year).
- End with a **bibliography** of both articles in APA format.
- Your **Turnitin score must be under 20%**. Points will be deducted if it is higher.

Using AI to draft

You may use NotebookLM to write a first draft with Prompt 6. A draft is a starting point, not a finished paper. You must substantially revise it so the final version is in your own words and reflects your own thinking.

> **Prompt 6. Draft a paragraph (NotebookLM)**
>
> I need to write one or two paragraphs, 150 to 350 words total, about the source [ARTICLE TITLE] for a class assignment. Before you write anything, interview me one question at a time about my future profession and why this article interests me. Ask no more than four questions.
>
> Then write a draft that explains why I chose the article, what the study did and found, and how the information will help me in my profession. Use plain language. Do not use any quotations, and do not copy phrases from the article. Paraphrase everything.

> **Why revising matters**
>
> AI summaries often reuse phrases from the article, and Turnitin counts those phrases as matches. Unrevised AI text is the fastest way to go over 20%.
>
> Turnitin also flags writing that looks AI generated. That score is separate from the similarity score, and your professor sees both.
>
> AI does not know why *you* chose the article or how *you* plan to use it. Only you can write that part well.

Revision checklist

- ☐ I rewrote each sentence in my own words instead of just swapping a few words.
- ☐ I added my own reasons for choosing the article and how it connects to my profession.
- ☐ I checked every fact and number in the draft against the article.
- ☐ There are no quotations, and no phrases copied from the article or the AI.
- ☐ Each write up is 150 to 350 words and ends with an APA in text citation.
- ☐ My bibliography lists both articles in APA format.

AI use statement

At the end of your paper, after the bibliography, add two or three sentences that say which AI tools you used and what you used them for. Add APA references for those tools to your bibliography. For example, “I used Gemini to brainstorm search terms and NotebookLM to summarize both articles and draft my first paragraph, which I then revised.”

Part 8. Reflection

This part is open ended. Write what you actually think. A short paragraph for each is plenty.

8a. What did AI help you do faster or better in this assignment? Where did it fall short?

8b. In your future profession, when would it be appropriate to use AI, and when would it not? Consider accuracy, patient privacy, and who is affected if the AI is wrong.

8c. Name one habit you will carry forward when you use AI for school or work.

Resource pages. Keep these for later

The sliding scale of subjectivity

More objective	In between	More subjective
What muscles make up the rotator cuff? How does glycolysis work? What does a physical therapist assistant do?	Which exercises work best for chronic low back pain? Is this study's sample large enough? Why do patients stop doing their home exercises?	Should health insurance cover chiropractic care? Should schools require vaccines for entry? The Israeli and Palestinian conflict
AI is usually reliable. Still verify specific numbers and citations.	**Be moderately critical.** Ask for evidence and for competing explanations.	**Treat the answer as one viewpoint among many.** Ask for several perspectives and read them yourself.

Numbers and citations are a special case. Even on objective topics, AI frequently gets statistics, dates, and sources wrong, and it can invent research articles that do not exist. Always check them against the original.

NotebookLM or Gemini?

	NotebookLM	Gemini
What it works from	Only the sources you upload, such as your two article PDFs	Its general training plus web searches
Strengths	Answers are tied to your sources, with numbered citations you can click to see the exact passage. Good for summarizing, comparing two articles, and finding the Methods and Results	Knows a little about almost everything. Good for building search strings for Library Search, explaining unfamiliar words like randomized controlled trial, checking APA formatting, and citing web pages
Weaknesses	Knows nothing beyond your sources. Can still misread or oversimplify a study. Its summaries often reuse the article's wording, which raises your Turnitin score	Much more likely to invent facts, statistics, and even whole articles that do not exist. Sounds just as confident when it is wrong
Use it when	You have the article and want to understand what it says	You do not have a source yet, or you need general help

Library Search quick guide

Library Search is the main search box on library.byui.edu. Use Gemini to build search strings and Library Search to find the articles.

Do this	Example	What it does
Put phrases in quotation marks	"low back pain"	Finds the words together and in that order
Group synonyms with OR in parentheses	(exercise OR "physical therapy")	Finds either word, so you get more results
Join different ideas with AND	"low back pain" AND exercise	Every result must include both ideas, so you get fewer results
Add an original research word	AND (trial OR participants)	Leans your results toward studies with their own data

- **Limiters.** Check **Peer Reviewed** and **Full Text** on the results page.
- **Too many results.** Add another idea with AND, or ask Gemini for a narrower string.
- **Too few results.** Remove an idea, add synonyms with OR, or ask Gemini for a broader string.
- **Tools on the article record.** Use **Cite** for an APA reference, **Export** to save to EndNote Fusion, and **Permalink** for a link that brings you back to the article.
- **Asked to pay for an article?** Go back to Library Search and open it from there, signed in with your campus account. Library Search links take you through the library's subscriptions.

Is it original research?

☐ The authors collected or analyzed their own data.

☐ It has Methods and Results sections.

☐ The abstract describes participants, a sample, or an intervention.

☐ The title does not include review, systematic review, meta analysis, or commentary.

☐ It was published in a peer reviewed journal.

Checklist for verifying AI output

☐ Ask the AI where each claim came from, and open the source yourself.

☐ Ask the AI what part of its answer it is least sure about.

☐ Check every number against the original article.

☐ Never trust an article title, author, or DOI that an AI gives you until you find the article in Library Search.

☐ Ask for the strongest opposing view on anything subjective.

☐ Compare with a second source that is not an AI.

Setting up EndNote Fusion

1. From the library website, find **EndNote Fusion** (it may still be listed as RefWorks) and create an account with your campus email.

2. Save articles to it. In Library Search, open the article record, click **Export**, and choose EndNote Fusion or RefWorks. You can also drag a PDF into EndNote Fusion and it will read the details.
3. Check each saved reference for missing or wrong details, just as you would check a cite tool.
4. Install the add in. In Word, go to **Insert** and then **Get Add ins** and search for EndNote Fusion or RefWorks Citation Manager. In Google Docs, look under **Extensions**.
5. Sign in to the add in, choose **APA 7th edition**, and insert citations as you write. The add in builds your bibliography automatically.

RefWorks was renamed EndNote Fusion in fall 2026, so some screens and menus may still show the old name. The older Write N Cite plugin no longer works. Use the add in instead.